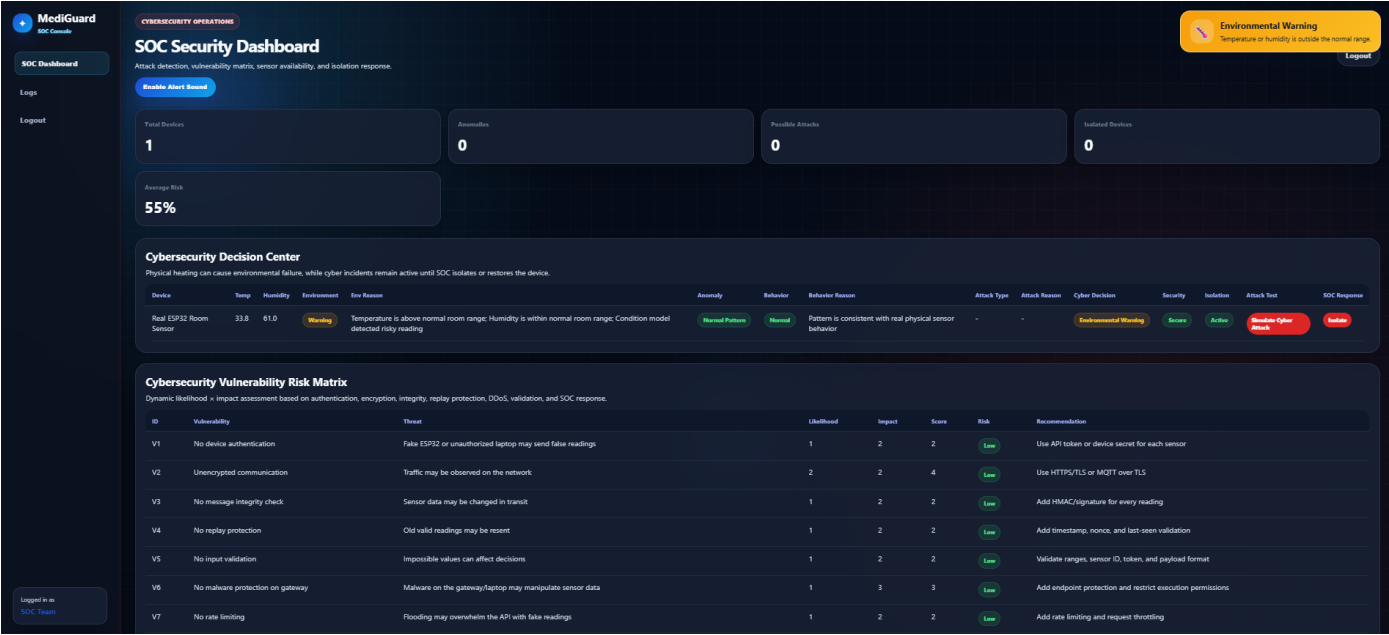
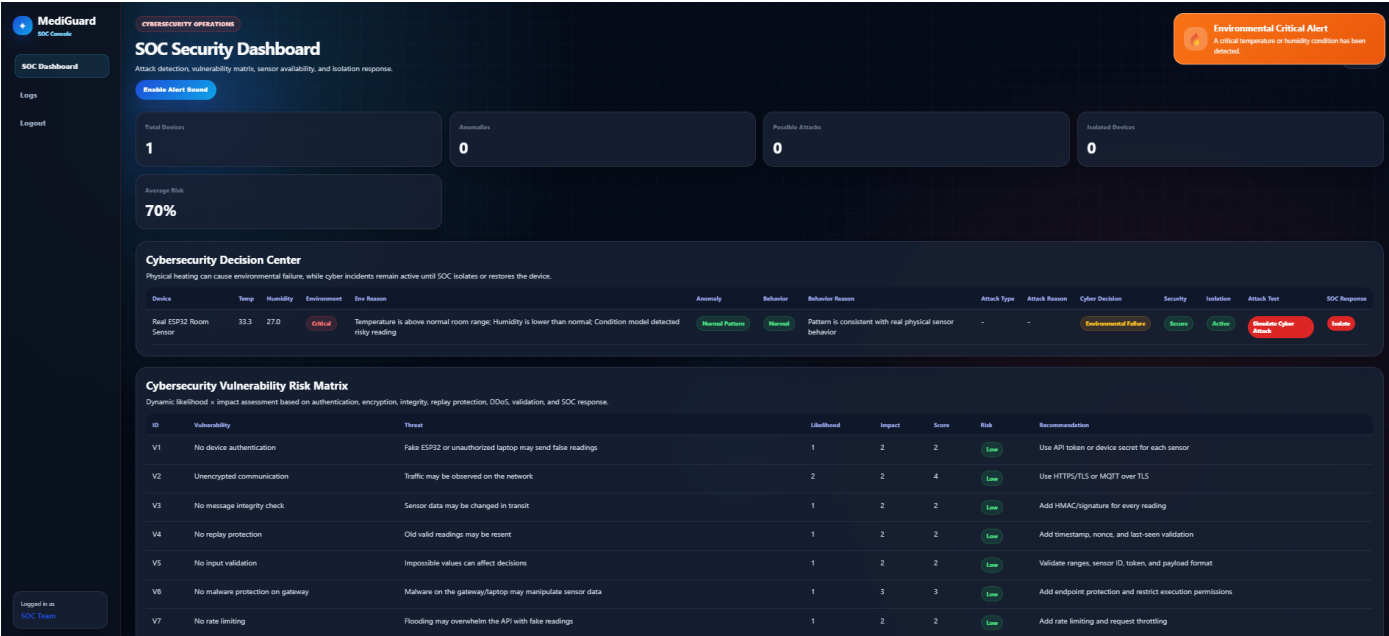


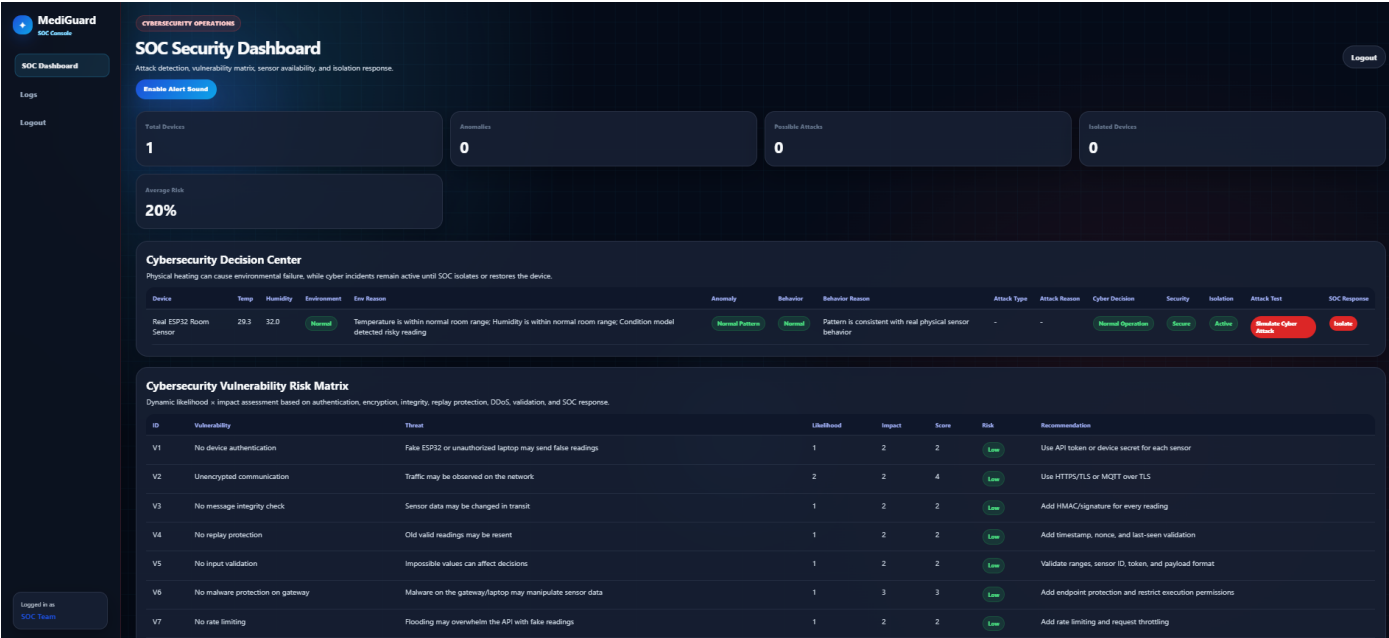
SOC Dashboard - Environmental Warning



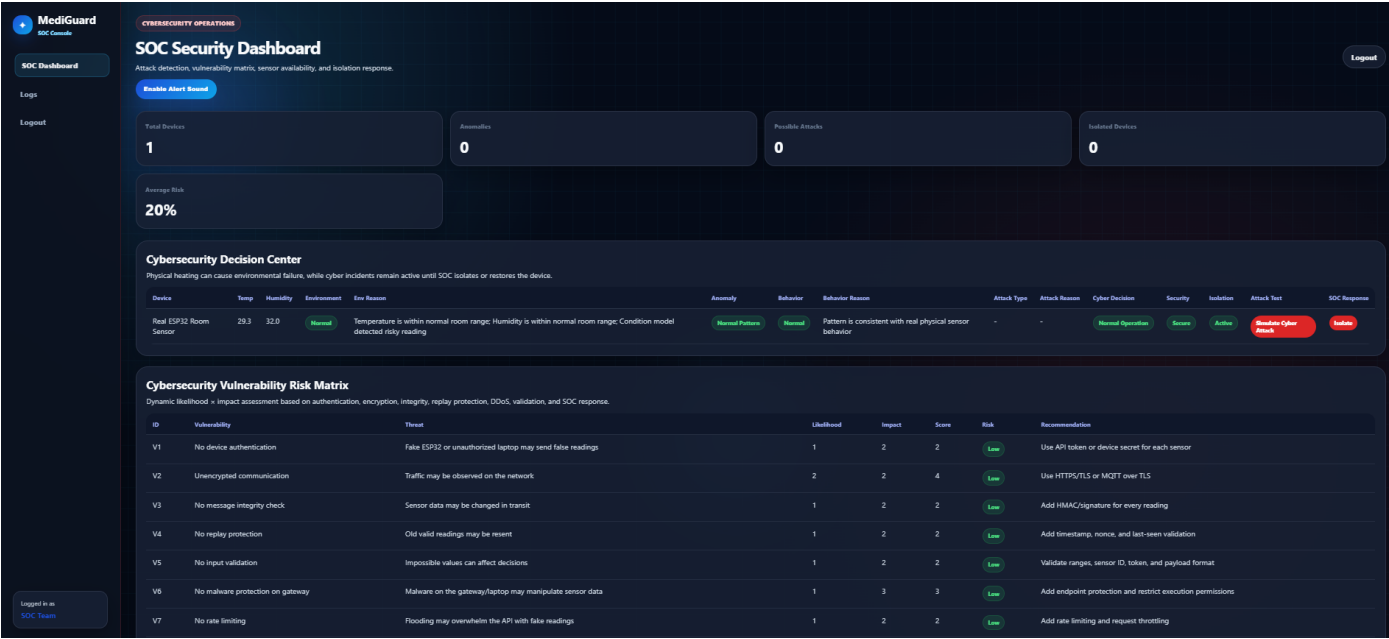
SOC Dashboard - Critical Environmental Alert



SOC Dashboard - Normal Monitoring State



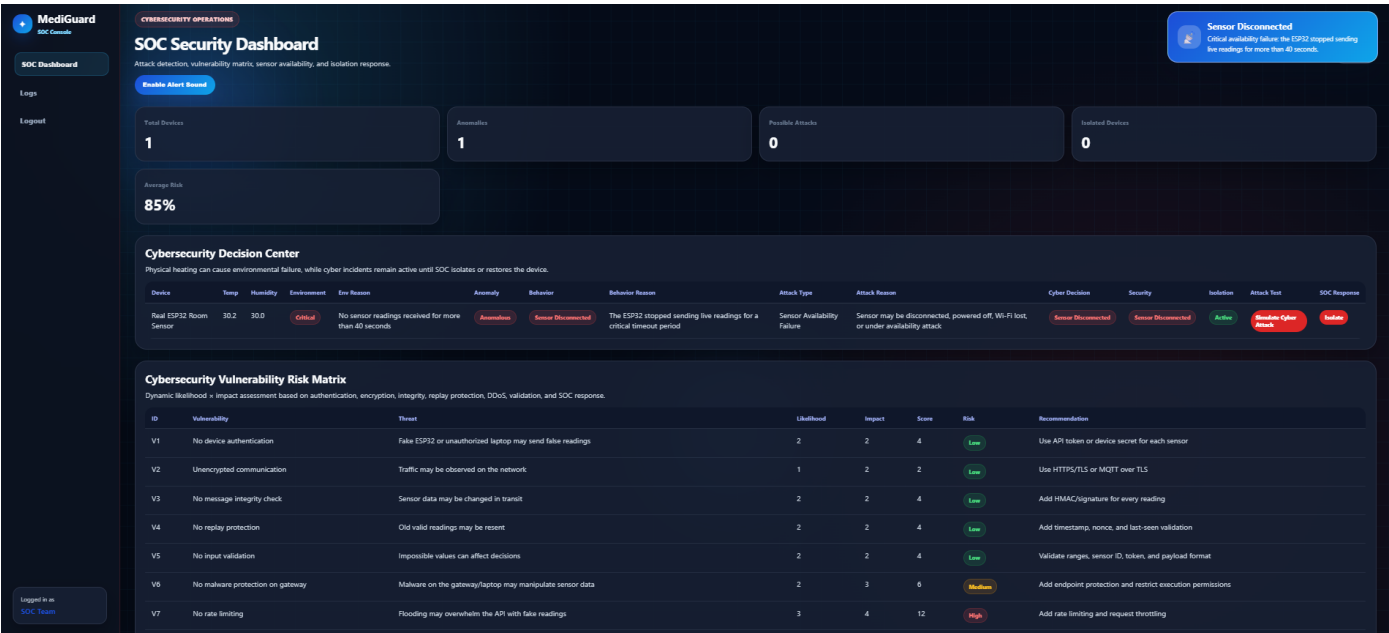
SOC Dashboard - Stable Monitoring State



MediGuard AI Landing Page



SOC Dashboard - Sensor Availability Failure



Cybersecurity Vulnerability Risk Matrix

Cybersecurity Vulnerability Risk Matrix

Dynamic likelihood × impact assessment based on authentication, encryption, integrity, replay protection, DDoS, validation, and SOC response.

ID	Vulnerability	Threat	Likelihood	Impact	Score	Risk	Recommendation
V1	No device authentication	Fake ESP32 or unauthorized laptop may send false readings	2	2	4	Low	Use API token or device secret for each sensor
V2	Unencrypted communication	Traffic may be observed on the network	1	2	2	Low	Use HTTPS/TLS or MQTT over TLS
V3	No message integrity check	Sensor data may be changed in transit	2	2	4	Low	Add HMAC/signature for every reading
V4	No replay protection	Old valid readings may be resent	2	2	4	Low	Add timestamp, nonce, and last-seen validation
V5	No input validation	Impossible values can affect decisions	2	2	4	Low	Validate ranges, sensor ID, token, and payload format
V6	No malware protection on gateway	Malware on the gateway/laptop may manipulate sensor data	2	3	6	Medium	Add endpoint protection and restrict execution permissions
V7	No rate limiting	Flooding may overwhelm the API with fake readings	3	4	12	High	Add rate limiting and request throttling
V8	No SOC isolation response	Compromised or unavailable sensor may remain trusted	4	4	16	High	Allow SOC to isolate suspicious sensors
V9	Isolated sensor detected	Sensor removed from trusted monitoring	1	1	1	Low	Inspect and restore only after validation
V10	No sensor availability monitoring	Sensor disconnection, Wi-Fi loss, or availability attack may go unnoticed	5	5	25	Critical	Track sensor heartbeat and last-seen timestamp